

Percolation, Random Walks, and Minimal Cutsets on Infinite Graphs

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- Compared the proofs of the paper’s theorems (bond percolation setting) to their site percolation analogues and identified asymmetries.
- Showed that the proof approach in both theorems is fundamentally not directly replicable.
- Partially extended previous results about the ‘gap at 1’ phenomenon for site p_c .

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- Discussion about the proof strategy in the case of site percolation and a counterexample in the setting of Theorem 1.

Let $\mathcal{L} = (V, E)$ be an infinite, connected graph. Fix $p \in [0, 1]$. In the *bond percolation* process with parameter p , each edge is independently open (included) with probability p and closed (removed) with probability $1 - p$.

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Denote the connected component of a vertex v in the subgraph of open edges by $C(v)$. The *percolation probability* at v is $\theta_v(p) := \mathbb{P}_p(|C(v)| = \infty)$ i.e the cluster of v is infinite in a bond percolation model with parameter p .

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The critical probability is $p_c(\mathcal{L}) := \sup\{p : \theta_v(p) = 0\}$, which is well-defined because it is independent of v . That is, positivity of the percolation probability is either a property of every vertex, or of none of them.

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By the monotonicity of the percolation probability, if p_c is non-trivial, there is a phase transition in the sense that:

$$\theta_v(p) \begin{cases} = 0 & \text{if } p < p_c, \\ > 0 & \text{if } p > p_c. \end{cases}$$

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Define $Q_n(v)$ to be the set of minimal cutsets from v to ∞ of size n . Set $q_n := \sup_{v \in V} |Q_n(v)|$ and $\kappa(G) := \sup_{n \geq 1} q_n^{1/n}$.

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Define the *uniform* critical probability as

$$p_c^*(G) := \inf\{p \in [0, 1] : \inf_{v \in V} \theta_v(p) > 0\},$$

which differs slightly from the usual critical parameter, in that the uniform version requires the percolation probability to be bounded away from zero over all vertices, not just that they are all positive. These notions coincide for graphs that are (quasi-) transitive.

The authors present two results, the first being a converse to the Peierls direction.

Theorem (Theorem 1, EST25)

For every infinite, connected, locally finite graph G :

$$p_c^*(G) < 1 \quad \Longleftrightarrow \quad \kappa(G) < \infty.$$

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Sketch proof of $\Rightarrow$. We want to show that

$$(\exists p < 1 \ \exists \theta > 0 \ \forall u \in V \ \mathbb{P}_p(u \leftrightarrow \infty) \geq \theta) \quad \Rightarrow \quad (\exists K < \infty \ \forall u \in V \ \forall n \geq 1 \ |Q_n(u)| \leq K^n)$$

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Fix $\theta, p \in (0, 1)$ such that $\mathbb{P}_p(u \leftrightarrow \infty) \geq \theta$ for every $u \in V$. Fix $o \in V$ and $n \geq 1$. Writing C for the cluster of o , we show that for every minimal cutset Π from o to ∞ with $|\Pi| = n$,

$$\mathbb{P}_p(\partial_\infty C = \Pi) \geq 1/K^n, \tag{1}$$

where $\partial_\infty C$ is the exposed edge-boundary of C , and $K = K(p, \theta) \in (0, \infty)$.

This concludes the proof since

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Any infinite open path from a vertex $u \in A$ must intersect B before exiting A , thus the hypothesis $\mathbb{P}_p(u \leftrightarrow \infty) \geq \theta$ implies that for all $u \in A$, $\mathbb{P}_p(u \overset{A}{\leftrightarrow} B) \geq \theta$.

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Let $\mathcal{E} = \{\forall b \in B, b \overset{A}{\leftrightarrow} o\}$ and let $\mathcal{F} = \{\text{all edges of } \Pi \text{ are closed}\}$. $\mathcal{F}$ and $\mathcal{E}$ are independent, so $\mathbb{P}_p(\mathcal{E} \cap \mathcal{F}) = \mathbb{P}_p(\mathcal{E})\mathbb{P}_p(\mathcal{F})$.

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If the event $\mathcal{E} \cap \mathcal{F}$ occurs, then the cluster C of o satisfies $B \subset C \subset A$.

Lemma (Lemma 2, EST25)

$$(B \subset C \subset A) \implies (\partial_\infty C = \Pi).$$

Hence, $\mathbb{P}_p(\partial_\infty C = \Pi) \geq \mathbb{P}_p(\mathcal{E} \cap \mathcal{F}) = \mathbb{P}_p(\mathcal{E})\mathbb{P}_p(\mathcal{F}) = \mathbb{P}_p(\mathcal{E})(1-p)^n$

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$$\mathbb{P}_p(\mathcal{E}) \geq c^{|B|} \geq c^n, \quad c = (p\theta/2)^{3/\theta}.$$

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The following theorem is the second theorem of the paper.

Theorem (Theorem 2, EST25)

Let G be an infinite, connected, locally finite graph. If G is uniformly transient, then $\kappa(G) < \infty$.

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Sketch proof. Prescribe exposed boundary and lower bound its probability using uniform transience, before summing over cutsets. $\square$

Extension of (Theorem 1, EST25) to Site Percolation

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Analogously, we call a set of vertices $W \subset V$ a vertex-cutset from a vertex v to ∞ if v belongs to a finite connected component of $(V \setminus W, E')$ where E' is the set of edges $uv \in E$ such that $u, v \in V \setminus W$.

Let $Q_n^{\text{site}}(v)$, q_n^{site} , $\kappa^{\text{site}}(G)$, and $p_c^{*\text{site}}(G)$ be defined analogously to the bond percolation definitions

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I claim that this is not true in general, and show this by counterexample.

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The idea of the proof was to impose events $\mathcal{E}$ and $\mathcal{F}$ such that $\{\mathcal{E} \cap \mathcal{F}\} \subset \{\partial_\infty^V C = \Pi\}$. If $\mathcal{E}$ is the event that every vertex in Π has a neighbour in A that is connected to o by an open path in A , and $\mathcal{F}$ is the event that every vertex of Π is closed, then $\{\mathcal{E} \cap \mathcal{F}\} = \{\partial_\infty^V C = \Pi\}$. Hence $\mathbb{P}_p^{\text{site}}(\partial_\infty^V C = \Pi) = \mathbb{P}_p^{\text{site}}(\mathcal{E})(1-p)^n$.

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Counterexample 1: Fix a constant $m > 1$, fix $p \in (0, 1)$, and let $L \geq 1$. Consider a path $B^{(L)} = (v_1, v_2, \dots, v_{mL})$ and let $B_i^{(L)} = (v_{(i-1)L+1}, v_{(i-1)L+2}, \dots, v_{iL})$ for all $i \in \{1, 2, \dots, m\}$. Let $o = v_1$ and note that for all $u \in B^{(L)}$, $\mathbb{P}_p^{\text{site}}(u \leftrightarrow B^{(L)}) = p$ since as long as the vertex u is open, it is contained in $B^{(L)}$. Moreover, the only route from o to $B_m^{(L)}$ is through all intermediate vertices in the path. Hence, given that o is open,

$$\mathbb{P}_p^{\text{site}}(\{o \leftrightarrow B_1^{(L)}\} \cap \dots \cap \{o \leftrightarrow B_m^{(L)}\}) \leq \mathbb{P}_p^{\text{site}}(\{o \leftrightarrow B_m^{(L)}\}) = p^{(m-1)L}.$$

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Let $H^{(L)} = B^{(L)} \cup \Pi^{(L)}$, where $\Pi^{(L)} = \{\pi_1, \dots, \pi_m\}$ and there is an edge between π_i and every vertex in $B_i^{(L)}$, for every $1 \leq i \leq m$.

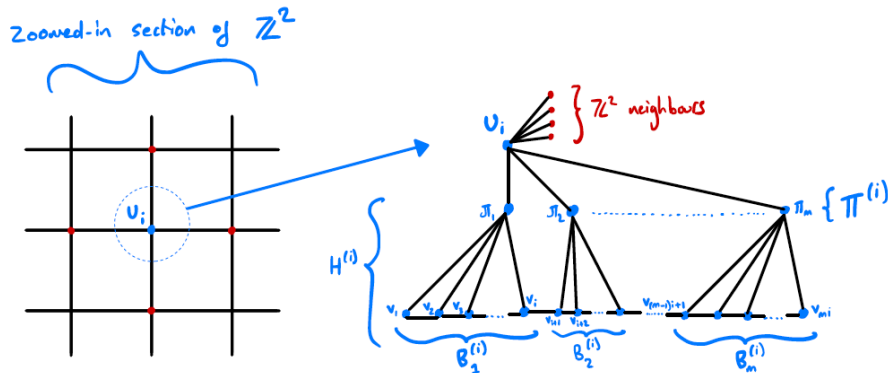
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We embed versions of this finite graph $H^{(L)}$ into $\mathbb{L}^2$. More precisely, enumerate the vertices of $\mathbb{L}^2$ and for the i^{th} element u_i in the enumeration, embed $H^{(i)}$ by simply adding edges between u_i and each of the vertices in $\Pi^{(i)}$.

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The following picture is an illustration:



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For any vertex u in this graph, $\mathbb{P}_p^{\text{site}}(u \leftrightarrow \infty) \geq p^2 \theta_{\mathbb{L}^2}^*(p)$. This is because u is either in $\mathbb{Z}^2$, is a Π -vertex which is the neighbour of a vertex in $\mathbb{Z}^2$, or is the neighbour of a Π -vertex. In any case, openness of at most two vertices means that u can reach the subgraph $\mathbb{L}^2$, and the inequality follows from the FKG inequality.

Therefore, for an appropriate choice of $p < 1$, this new graph can inherit the non-triviality of the uniform percolation probability $\theta_{\mathbb{L}^2}^*(p)$, and thus also has $p_c^{\text{site}} < 1$, satisfying the hypothesis of the theorem.

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Finally, by the construction of the graph, we can find an origin vertex o (at the start of a path in an embedded graph) and minimal cutset Π of length m such that $\mathbb{P}_p^{\text{site}}(\partial_\infty^V C(o) = \Pi) < 1/K^m$ for any constant $K(p) > 0$.

Indeed, for all $L \geq 1$ we can find an embedded graph with path of length mL , giving that $\mathbb{P}_p^{\text{site}}(\mathcal{E}) \leq p^{(m-1)L}$, and we can take L large enough.

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